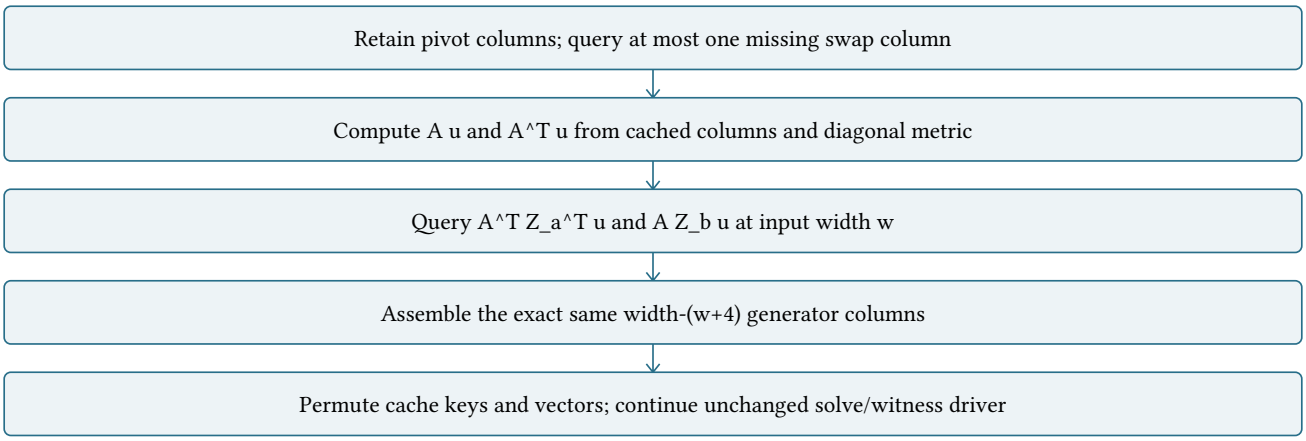
Paired weighted-symmetry transport

2026-11-02 · Derived identical-generator update · Numerical trials: zero

Identity. For $u=e_i-e_j$ and $P=I-uu^T$, compose the existing row then column transport. The exact generator columns for PAP are:

$$G' = [PG,\ u,\ -Z_a u,\ PAu,\ -PAZ_\beta u] \quad K' = [PK,\ PA^T Z_a^T u,\ PA^T u,\ Z_\beta^T u,\ u]$$



Reuse. With $c_i=Ae_i$ and $c_j=Ae_j$, $Au=c_i-c_j$ and $A^Tu=D(d_i^{-1}c_i-d_j^{-1}c_j)$. The selected pivot column is always cached. Only the other column may be missing. After swapping, cache key t becomes P(t), and its vector becomes Pc_t. This preserves provenance across pair pivots.

Derived per nontrivial swap. Let h be missing columns (0 or 1) and w input width. Identity swaps remain zero-query controls.

Quantity	Sequential reference	Paired transport
Source applications	4	2+h
Query widths	w,w,w+2,w+2	w throughout
Transpose / forward	2 / 2	1 / (1+h)
Output width	w+4	w+4
Extra field products	4	3n+4
Extra field additions	2n+2	4n+1

Native-product accounting. The preserved fused kernel records 4w+8 native products per successful application. Reference total: 16w+48. Paired total: (2+h)(4w+8). Savings are 4w+24 for h=1 or 8w+32 for h=0, alongside 3n additional field products and 2n−1 additions. These units are distinct. Bit lengths, cache copies and runtime are unmeasured. For a second swap use its actual input width w+4.

Integration and ownership. Only the complete driver's transport import changes. Final G′/K′ columns match the reference in ordering and signs, under the stated identities. Cache starts with one/two selected columns and grows by at most one per swap (at most four retained). Selection is one-use and source-bound; unchanged generators and internal metric/cache provenance are required. Query/event paths and counters intentionally differ.

Checks / next. Full driver normalized-AST equality and dependency hashes; all Python files syntax checked. No numerical, mutation, heap or performance trial. Validate exact generator arrays, all swap positions, both cache branches, pair reuse, nonuniform metrics, corruptions, ledger widths and full root solution/witness equations under the unchanged upstream cohort. Next derive metric provenance views and safe zero-prefix reuse; neither saving is counted.

Sources. paired.py, driver_paired.py, check_preparation.py, DERIVATION.md, and preserved row/column transport modules. Bostan et al. (2017), equation (2) and Section 1, DOI 10.1137/16M1062855, provides established shift/displacement context; no new displacement theory or measured speedup is claimed.